

statprojectscript.R

coolm

2025-10-28

```
#Sion King  
#Stat 3 Project 1
```

```
#Question 2  
head(grad_data, 10)
```

```
##   GRE TOEFL Urate SOP LOR CGPA Chance  
## 1  337   118    4 4.5 4.5 3.86    92  
## 2  324   107    4 4.0 4.5 3.55    76  
## 3  316   104    3 3.0 3.5 3.20    72  
## 4  322   110    3 3.5 2.5 3.47    80  
## 5  314   103    2 2.0 3.0 3.28    65  
## 6  330   115    5 4.5 3.0 3.74    90  
## 7  321   109    3 3.0 4.0 3.28    75  
## 8  308   101    2 3.0 4.0 3.16    68  
## 9  302   102    1 2.0 1.5 3.20    50  
## 10 323   108    3 3.5 3.0 3.44    45
```

```
#Question 3  
summary(grad_data[, c("GRE", "CGPA", "Chance")])
```

```
##           GRE           CGPA           Chance  
## Min.   :290.0   Min.   :2.720   Min.   :34.00  
## 1st Qu.:308.0   1st Qu.:3.270   1st Qu.:64.00  
## Median :317.0   Median :3.445   Median :73.00  
## Mean   :316.8   Mean   :3.440   Mean   :72.44  
## 3rd Qu.:325.0   3rd Qu.:3.623   3rd Qu.:83.00  
## Max.   :340.0   Max.   :3.970   Max.   :97.00
```

```
par(mfrow = c(2, 2))
```

```
#Question 4  
hist(grad_data$Chance,  
     main = "Histogram of Chance of Admission",  
     xlab = "Chance of Admission (%)",  
     col = "lightpink")
```

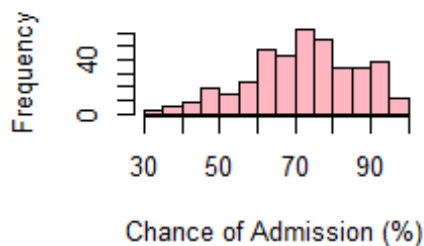
```
hist(grad_data$CGPA,  
     main = "Histogram of CGPA",  
     xlab = "CGPA (out of 4)",  
     col = "lightblue")
```

```
hist(grad_data$GRE,
```

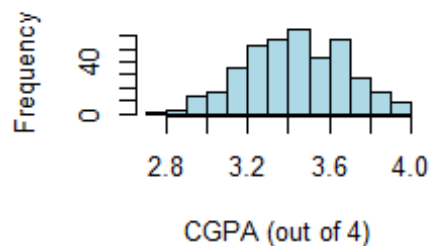
```
main = "Histogram of GRE Score",
xlab = "GRE Score (out of 340)",
col = "lightyellow")
```

```
hist(grad_data$TOEFL,
main = "Histogram of TOEFL Score",
xlab = "TOEFL Score (out of 120)",
col = "lightgreen")
```

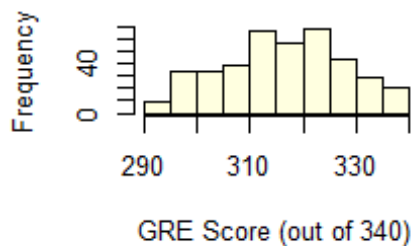
Histogram of Chance of Admiss



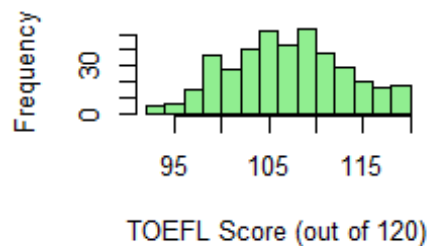
Histogram of CGPA



Histogram of GRE Score



Histogram of TOEFL Score



#Question 4:

#Chance Histogram: Appears left skewed. Not normally distributed. The skewing of the data means students in this dataset are more likely to be accepted than rejected

#CGPA Histogram: Appears approximately normally distributed. If any skewness, very slight left skewness. A large proportion of student's CGPA in this dataset is between 3.2 and 3.6 on average.

#GRE Score Histogram: Appears mostly symmetric (Unimodal). Approximately normally distributed, resembling a bell curve.

#TOEFL Score Histogram: Appears mostly symmetric (Unimodal). Approximately normally distributed, resembling a bell curve.

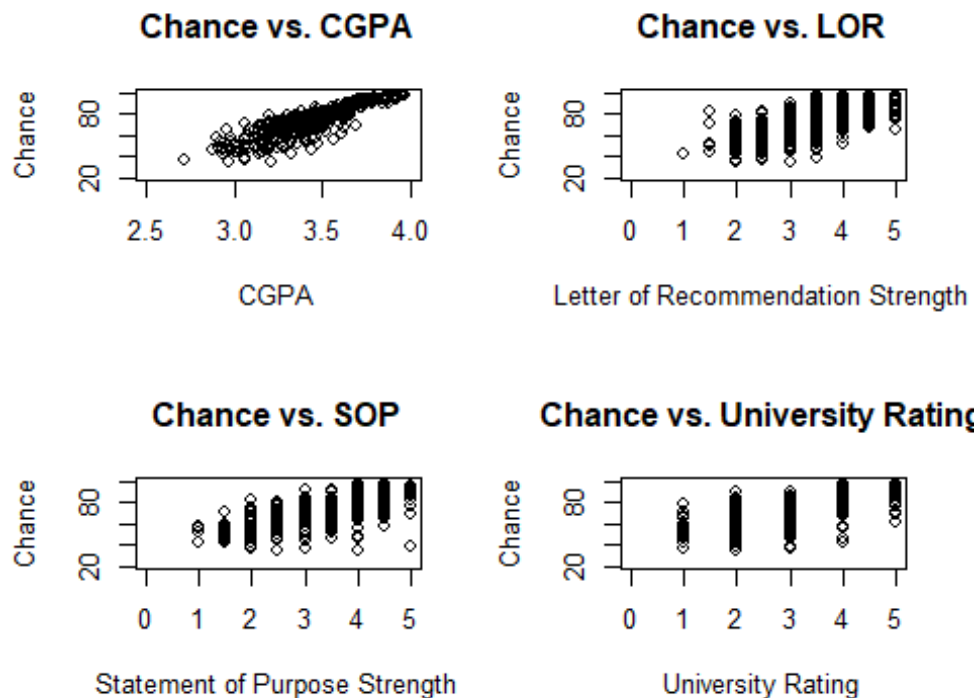
```
plot (x=grad_data$CGPA,y=grad_data$Chance,main="Chance vs. CGPA",
xlab="CGPA",ylab="Chance",xlim=c(2.5,4),ylim=c(20,100))
```

```
plot (x=grad_data$LOR,y=grad_data$Chance,main="Chance vs. LOR",
xlab="Letter of Recommendation")
```

```
Strength",ylab="Chance",xlim=c(0,5),ylim=c(20,100))

plot (x=grad_data$SOP,y=grad_data$Chance,main="Chance vs. SOP",
      xlab="Statement of Purpose
Strength",ylab="Chance",xlim=c(0,5),ylim=c(20,100))

plot (x=grad_data$Urate,y=grad_data$Chance,main="Chance vs. University
Rating",
      xlab="University Rating",ylab="Chance",xlim=c(0,5),ylim=c(20,100))
```



```
plot (x=grad_data$TOEFL,y=grad_data$Chance,main="Chance vs. TOEFL Scores",
      xlab="TOEFL Scores",ylab="Chance",xlim=c(90,120),ylim=c(20,100))

plot (x=grad_data$GRE,y=grad_data$Chance,main="Chance vs. GRE Scores",
      xlab="GRE Scores",ylab="Chance",xlim=c(285,345),ylim=c(20,100))
```

#Question 6:

#The variables of LOR, SOP, and University Rating in relation to CGPA have a positive relationship but are all nonlinear.

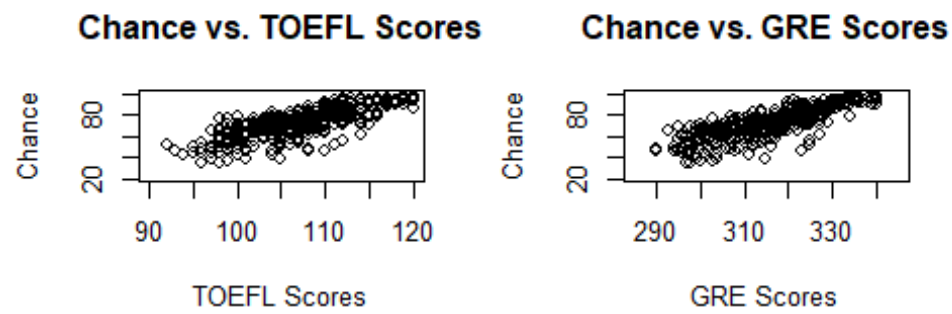
#The variables CGPA, TOEFL, and GRE in relation to CGPA are all positive linear relationships.

Question 7:

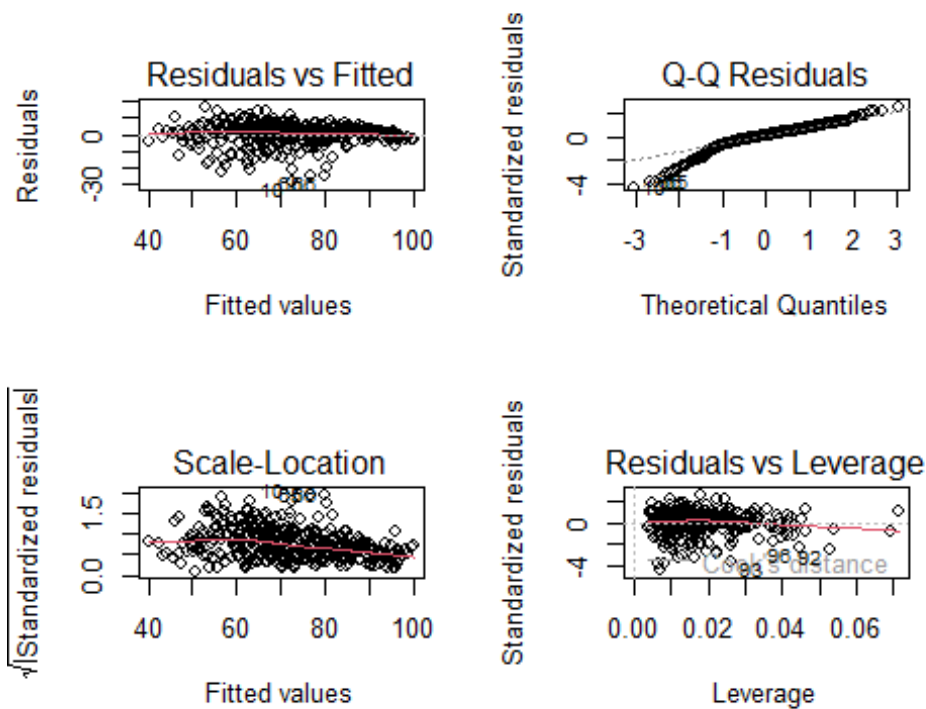
#ALL of the graphs show a positive linear trend, although LOR, SOP,

#and university rating are not graphed the same due to their distinct values

```
par(mfrow = c(2, 2))
```



```
plot(full_model)
```



```
par(mfrow = c(1, 1))
```

#The homoscedasticity assumption seems mostly valid, though a more definitive check

#would require a plot of the model's residuals versus the predicted values.

#We can confirm the validity of the independence assumption that the residuals are independent from the given graphs.

#We can confirm the validity of the linearity assumption from the given graphs. It is well-met by the residual graph

#We can confirm the validity of the normality assumption from the given graphs. It is well-met by the Q-Q plot.

#We can confirm the validity of the homoscedasticity assumption from the given graphs. It is well-met by the scale-location graph.

#We can confirm the validity of the influential outliers from the Residual vs Leverage graph. There are no outliers which have extreme influence.

#Question 8:

#The three best independent variables are CGPA, GRE Scores, and TOEFL Scores

#The model is $\text{Chance} = \beta_0 + \beta_1(\text{CGPA}) + \beta_2(\text{GRE}) + \beta_3(\text{TOEFL}) + \varepsilon$

#Question 9:

```
print(round(correlations_with_chance, 3))
```

```
##      GRE  TOEFL  Urate    SOP    LOR    CGPA  Chance
##  0.803  0.792  0.711  0.676  0.670  0.873  1.000
```

#From the correlation coefficients, I can deduce that, CGPA, GRE and TOEFL Scores

#have the strongest linear relationship with chance of admission due to their #high correlation coefficient values.

#The correlation coefficient is the strength and direction of a linear relationship

#The slope is the steepness of that relationship, telling you how much the y-value will change

#for a single unit increase in X

#Using both the correlation coefficient and the slope are important to conclude

#the significance of multiple linear regressions

#Question 10:

```
cat("\nCorrelation Hypothesis Tests for Each Independent Variable vs.
Chance\n\n")
```

```
##
```

```
## Correlation Hypothesis Tests for Each Independent Variable vs. Chance
```

Loop through each predictor and perform a correlation test against the response variable

```
for (independent_var in independent_vars) {
  cat(paste("Test for:", independent_var, "\n"))
  test_result <- cor.test(grad_data[[independent_var]],
grad_data[[dependent_var]])
  print(test_result)
  cat("\n")
}
```

```
## Test for: GRE
```

```
##
```

```
## Pearson's product-moment correlation
```

```
##
```

```
## data: grad_data[[independent_var]] and grad_data[[dependent_var]]
```

```
## t = 26.843, df = 398, p-value < 2.2e-16
```

```
## alternative hypothesis: true correlation is not equal to 0
```

```
## 95 percent confidence interval:
```

```
## 0.7647419 0.8349536
```

```
## sample estimates:
```

```
## cor
```

```
## 0.8026105
```

```
##
```

```
##
```

```
## Test for: TOEFL
##
## Pearson's product-moment correlation
##
## data: grad_data[[independent_var]] and grad_data[[dependent_var]]
## t = 25.845, df = 398, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.7519028 0.8255675
## sample estimates:
## cor
## 0.791594
##
##
## Test for: Urate
##
## Pearson's product-moment correlation
##
## data: grad_data[[independent_var]] and grad_data[[dependent_var]]
## t = 20.186, df = 398, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.6591685 0.7565413
## sample estimates:
## cor
## 0.7112503
##
##
## Test for: SOP
##
## Pearson's product-moment correlation
##
## data: grad_data[[independent_var]] and grad_data[[dependent_var]]
## t = 18.288, df = 398, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.6186712 0.7257010
## sample estimates:
## cor
## 0.6757319
##
##
## Test for: LOR
##
## Pearson's product-moment correlation
##
## data: grad_data[[independent_var]] and grad_data[[dependent_var]]
## t = 18, df = 398, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
```

```

## 0.6120380 0.7206083
## sample estimates:
##      cor
## 0.6698888
##
## Test for: CGPA
##
## Pearson's product-moment correlation
##
## data: grad_data[[independent_var]] and grad_data[[dependent_var]]
## t = 35.776, df = 398, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.8479462 0.8948062
## sample estimates:
##      cor
## 0.8733827

#Question 11:

cat("\n Univariate Linear Regression Model Summaries \n\n")

##
## Univariate Linear Regression Model Summaries

# Loop through each predictor to create and summarize a simple linear model
for (independent_var in independent_vars) {
  # Create a formula for the linear model, e.g., Chance ~ GRE
  formula <- as.formula(paste(dependent_var, "~", independent_var))

  # Fit the linear model
  model <- lm(formula, data = grad_data)

  # Print the summary which includes the hypothesis test for the slope
  cat(paste(" Model Summary for:", independent_var, "\n"))
  print(summary(model))
  cat("\n")
}

## Model Summary for: GRE
##
## Call:
## lm(formula = formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -33.613  -4.604   0.408   5.644  18.339
##
## Coefficients:

```



```

##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -243.60842    11.78141  -20.68  <2e-16 ***
## GRE          0.99759     0.03716   26.84  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.517 on 398 degrees of freedom
## Multiple R-squared:  0.6442, Adjusted R-squared:  0.6433
## F-statistic: 720.6 on 1 and 398 DF,  p-value: < 2.2e-16
##
##
## Model Summary for: TOEFL
##
## Call:
## lm(formula = formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.252  -5.128   1.328   5.453  21.067
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -127.34005     7.74217  -16.45  <2e-16 ***
## TOEFL        1.85993     0.07197   25.84  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.725 on 398 degrees of freedom
## Multiple R-squared:  0.6266, Adjusted R-squared:  0.6257
## F-statistic: 667.9 on 1 and 398 DF,  p-value: < 2.2e-16
##
##
## Model Summary for: Urate
##
## Call:
## lm(formula = formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -38.527  -4.560   1.473   6.341  27.209
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  45.0537     1.4463   31.15  <2e-16 ***
## Urate        8.8684     0.4393   20.19  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.04 on 398 degrees of freedom
## Multiple R-squared:  0.5059, Adjusted R-squared:  0.5046

```

```

## F-statistic: 407.5 on 1 and 398 DF,  p-value: < 2.2e-16
##
##
## Model Summary for: SOP
##
## Call:
## lm(formula = formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.748  -5.392   1.823   7.037  22.393
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  39.8942     1.8556   21.50  <2e-16 ***
## SOP          9.5708     0.5233   18.29  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.53 on 398 degrees of freedom
## Multiple R-squared:  0.4566, Adjusted R-squared:  0.4552
## F-statistic: 334.4 on 1 and 398 DF,  p-value: < 2.2e-16
##
##
## Model Summary for: LOR
##
## Call:
## lm(formula = formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -34.940  -6.256   0.060   7.389  29.325
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  35.7256     2.1072   16.95  <2e-16 ***
## LOR          10.6327     0.5907   18.00  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.6 on 398 degrees of freedom
## Multiple R-squared:  0.4488, Adjusted R-squared:  0.4474
## F-statistic:  324 on 1 and 398 DF,  p-value: < 2.2e-16
##
##
## Model Summary for: CGPA
##
## Call:
## lm(formula = formula, data = grad_data)
##

```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -27.4598  -2.9585   0.9733   4.1799  18.0757
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -107.215      5.034  -21.30  <2e-16 ***
## CGPA         52.231      1.460   35.78  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.954 on 398 degrees of freedom
## Multiple R-squared:  0.7628, Adjusted R-squared:  0.7622
## F-statistic: 1280 on 1 and 398 DF,  p-value: < 2.2e-16
```

#Question 12:

```
cat("\n ANOVA Tables for Top Two Models by R-squared \n\n")

##
## ANOVA Tables for Top Two Models by R-squared

r_squared_values <- c()
# Calculate R-squared for each univariate model
for (independent_var in independent_vars) {
  formula <- as.formula(paste(dependent_var, "~", independent_var))
  model <- lm(formula, data = grad_data)
  r_squared_values[independent_var] <- summary(model)$r.squared
}

# Sort the predictors by R-squared value in descending order and get the top two
top_two_independent_vars <- names(sort(r_squared_values, decreasing =
TRUE)[1:2])

cat(paste("The two independent variables with the highest R-squared values
are:",
          top_two_independent_vars[1], "and", top_two_independent_vars[2],
          "\n\n"))

## The two independent variables with the highest R-squared values are: CGPA
and GRE

# Generate and print the ANOVA table for the top two models
for (independent_var in top_two_independent_vars) {
  formula <- as.formula(paste(dependent_var, "~", independent_var))
  model <- lm(formula, data = grad_data)

  cat(paste("ANOVA Table for Model with", independent_var, "\n"))
  print(anova(model))
}
```

```

cat("\n")
}

## ANOVA Table for Model with CGPA
## Analysis of Variance Table
##
## Response: Chance
##           Df Sum Sq Mean Sq F value    Pr(>F)
## CGPA         1  61898    61898  1279.9 < 2.2e-16 ***
## Residuals   398  19248         48
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ANOVA Table for Model with GRE
## Analysis of Variance Table
##
## Response: Chance
##           Df Sum Sq Mean Sq F value    Pr(>F)
## GRE          1  52273    52273   720.55 < 2.2e-16 ***
## Residuals   398  28873         73
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#Question 13:
cat("--- Full Multiple Linear Regression Model ---\n\n")

## --- Full Multiple Linear Regression Model ---

full_formula <- as.formula(paste(dependent_var, "~", paste(independent_vars,
collapse = " + ")))

full_model <- lm(full_formula, data = grad_data)

print(summary(full_model))

##
## Call:
## lm(formula = full_formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -27.9145  -2.3808   0.9351   3.5774  16.0571
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -141.4185    11.5420  -12.252 < 2e-16 ***
## GRE           0.2270     0.0578   3.928 0.000101 ***
## TOEFL         0.2762     0.1099   2.512 0.012398 *
## Urate         0.5995     0.4821   1.244 0.214367
## SOP          -0.2000     0.5604  -0.357 0.721336
## LOR           2.2784     0.5598   4.070 5.68e-05 ***

```

```
## CGPA          30.0138      3.0886   9.718 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.446 on 393 degrees of freedom
## Multiple R-squared:  0.7988, Adjusted R-squared:  0.7957
## F-statistic: 260 on 6 and 393 DF, p-value: < 2.2e-16

#Based on the p-values (Pr(>|t|)), the 'Urate' and 'SOP' variables are not
significant at the alpha = 0.05 level, as their p-values are greater than
0.05.

#Question 14:

cat("--- Reduced Multiple Linear Regression Model (Final Model) ---\n\n")
## --- Reduced Multiple Linear Regression Model (Final Model) ---

significant_predictors <- c("GRE", "TOEFL", "LOR", "CGPA")
reduced_formula <- as.formula(paste(dependent_var, "~",
paste(significant_predictors, collapse = " + ")))
# Fit the reduced model
final_model <- lm(reduced_formula, data = grad_data)

# Print the summary of the reduced model
cat("--- Summary of the Final Model ---\n")

## --- Summary of the Final Model ---

print(summary(final_model))

##
## Call:
## lm(formula = reduced_formula, data = grad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -27.9694  -2.2884   0.9324   3.6508  16.1704
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -146.25106    10.57306  -13.832 < 2e-16 ***
## GRE           0.23112     0.05762   4.011 7.23e-05 ***
## TOEFL         0.29291     0.10756   2.723 0.00675 **
## LOR           2.39604     0.48392   4.951 1.09e-06 ***
## CGPA         30.74029     2.96215  10.378 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.442 on 395 degrees of freedom
```

```
## Multiple R-squared:  0.798, Adjusted R-squared:  0.7959
## F-statistic: 390.1 on 4 and 395 DF,  p-value: < 2.2e-16

# Print ANOVA tables for comparison
cat("\n--- ANOVA Table for the Full Model ---\n")

##
## --- ANOVA Table for the Full Model ---

print(anova(full_model))

## Analysis of Variance Table
##
## Response: Chance
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## GRE         1  52273   52273 1258.162 < 2.2e-16 ***
## TOEFL        1   3921    3921   94.377 < 2.2e-16 ***
## Urate        1   2370    2370   57.032 3.030e-13 ***
## SOP          1    757     757   18.227 2.461e-05 ***
## LOR          1   1574    1574   37.880 1.857e-09 ***
## CGPA         1   3923    3923   94.433 < 2.2e-16 ***
## Residuals  393  16328      42
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("\n--- ANOVA Table for the Final (Reduced) Model ---\n")
```

```
##
## --- ANOVA Table for the Final (Reduced) Model ---

print(anova(final_model))

## Analysis of Variance Table
##
## Response: Chance
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## GRE         1  52273   52273 1259.606 < 2.2e-16 ***
## TOEFL        1   3921    3921   94.486 < 2.2e-16 ***
## LOR          1   4090    4090   98.565 < 2.2e-16 ***
## CGPA         1   4469    4469  107.696 < 2.2e-16 ***
## Residuals  395  16392      41
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#By removing the non-significant variables, the Residual Sum of Squares increased slightly, which is expected. However, the model is simpler while retaining nearly the same explanatory power. The F-statistic and overall model significance remain very high.

#Question 15:

```
cat("--- Confidence and Prediction Intervals for the Final Model ---\n\n")
```

```
## --- Confidence and Prediction Intervals for the Final Model ---

conf_intervals <- predict(final_model, interval = "confidence", level = 0.95)
pred_intervals <- predict(final_model, interval = "prediction", level = 0.95)

## Warning in predict.lm(final_model, interval = "prediction", level = 0.95):
## predictions on current data refer to _future_ responses

results_table <- data.frame(
  Actual_Chance = grad_data$Chance,
  Fitted_Value = conf_intervals[, "fit"],
  Conf_Lower_Bound = conf_intervals[, "lwr"],
  Conf_Upper_Bound = conf_intervals[, "upr"],
  Pred_Lower_Bound = pred_intervals[, "lwr"],
  Pred_Upper_Bound = pred_intervals[, "upr"]
)

print(head(results_table, 10))
```

```
##      Actual_Chance Fitted_Value Conf_Lower_Bound Conf_Upper_Bound
## 1             92      95.63917      94.29855      96.97979
## 2             76      79.88311      78.59288      81.17335
## 3             72      64.00029      62.59078      65.40979
## 4             80      73.04831      71.79269      74.30392
## 5             65      64.50634      63.56794      65.44474
## 6             90      85.85971      84.29827      87.42115
## 7             75      70.27768      68.62735      71.92801
## 8             68      61.24101      59.65670      62.82532
## 9             50      55.38672      53.71857      57.05487
## 10            45      72.96941      71.96779      73.97104
##      Pred_Lower_Bound Pred_Upper_Bound
## 1             82.90349             108.37486
## 2             67.15263              92.61359
## 3             51.25717              76.74341
## 4             60.32129              85.77532
## 5             51.80669              77.20599
## 6             73.09889              98.62053
## 7             57.50568              83.04968
## 8             48.47737              74.00465
## 9             42.61241              68.16104
## 10            60.26494              85.67389
```

#Prints the first 10 rows of the confidence and prediction intervals, all of the values are stored in the environment dataset

#Question 16:

#Chance = $-146.251 + 0.231\text{GRE} + 0.293\text{TOEFL} + 2.396\text{LOR} + 30.740\text{CGPA}$

#The intercept has no practical meaning, as it is impossible for someone to have a 0 all the variables. It is used to ensure the line of best fit is accurate.

#Each slope represents the average change in the Chance of admission for a

one-unit increase in its corresponding variable, assuming all other variables in the model are held constant.

#Question 17:

#Descriptive Statistics (like correlations and histograms) are perfect for the initial phase of exploration. They allowed us to understand the shape of our data (histogram/scatterplot) and get a "first look" at potential relationships.

#Regression and Regression Analysis is the necessary second phase for confirming, quantifying, and modeling those relationships.

#Regression Analysis provides a deeper look into the data than descriptive statistics does. Allowing both the project creator and viewers to understand the data more thoroughly

#I would recommend to include not only simple correlations, but also build a multiple regression model and check for variable significance.

#Building a correlation matrix, constructing a full model, and another final reduced model is what I recommend.